

```
from google.colab import files
uploaded = files.upload()
```

 kfood_dataset.zip

kfood_dataset.zip(application/x-zip-compressed) - 207825962 bytes, last modified: 16/11/2025 - 100% done
Saving kfood_dataset.zip to kfood_dataset.zip

```
import zipfile
import os

zip_path = 'kfood_dataset.zip' # ganti sesuai nama file kamu

with zipfile.ZipFile(zip_path, 'r') as zip_ref:
    zip_ref.extractall('/content/korean_food_dataset')

print("Selesai diekstrak!")
```

Selesai diekstrak!

```
!ls /content/korean_food_dataset
```

content

```
!ls /content/korean_food_dataset/content
```

korean_food_dataset

```
dataset_path = "/content/korean_food_dataset/content/korean_food_dataset"
```

```
from google.colab import files
uploaded = files.upload()
```

 efficientnet_korean_food.keras

efficientnet_korean_food.keras(n/a) - 130754900 bytes, last modified: 17/11/2025 - 100% done
Saving efficientnet_korean_food.keras to efficientnet_korean_food.keras

```
import tensorflow as tf
from tensorflow.keras.optimizers import Adam
from tensorflow.keras.models import load_model
from tensorflow.keras.preprocessing.image import ImageDataGenerator
```

```
model = load_model("efficientnet_korean_food.keras", compile=False)
print("Model loaded!")
```

Model loaded!

```
from tensorflow.keras.models import load_model

model = load_model("efficientnet_korean_food.keras")
```

```
import tensorflow as tf
from tensorflow.keras.models import load_model

model = load_model("efficientnet_korean_food.keras")

dataset_path = "/content/korean_food_dataset/content/korean_food_dataset"

ds = tf.keras.utils.image_dataset_from_directory(
    dataset_path,
    image_size=(300, 300),
    batch_size=32,
    label_mode="categorical",
    shuffle=False
)

loss, acc = model.evaluate(ds)
print("Akurasi keseluruhan:", acc)
```

Found 5000 files belonging to 25 classes.
157/157 ————— **39s** 157ms/step - accuracy: 0.0697 - loss: 627875.1875
Akurasi keseluruhan: 0.039400000125169754

```
model = load_model("efficientnet_korean_food.keras")
model.summary()
```


Model: "functional"

Layer (type)	Output Shape	Param #	Connected to
input_layer (InputLayer)	(None, 300, 300, 3)	0	-
rescaling (Rescaling)	(None, 300, 300, 3)	0	input_layer[0][0]
normalization (Normalization)	(None, 300, 300, 3)	7	rescaling[0][0]
rescaling_1 (Rescaling)	(None, 300, 300, 3)	0	normalization[0]...
stem_conv_pad (ZeroPadding2D)	(None, 301, 301, 3)	0	rescaling_1[0][0]
stem_conv (Conv2D)	(None, 150, 150, 40)	1,080	stem_conv_pad[0]...
stem_bn (BatchNormalizatio...	(None, 150, 150, 40)	160	stem_conv[0][0]
stem_activation (Activation)	(None, 150, 150, 40)	0	stem_bn[0][0]
block1a_dwconv (DepthwiseConv2D)	(None, 150, 150, 40)	360	stem_activation[...
block1a_bn (BatchNormalizatio...	(None, 150, 150, 40)	160	block1a_dwconv[0...
block1a_activation (Activation)	(None, 150, 150, 40)	0	block1a_bn[0][0]
block1a_se_squeeze (GlobalAveragePool...	(None, 40)	0	block1a_activati...
block1a_se_reshape (Reshape)	(None, 1, 1, 40)	0	block1a_se_squee...
block1a_se_reduce (Conv2D)	(None, 1, 1, 10)	410	block1a_se_resha...
block1a_se_expand (Conv2D)	(None, 1, 1, 40)	440	block1a_se_reduc...
block1a_se_excite (Multiply)	(None, 150, 150, 40)	0	block1a_activati... block1a_se_expan...
block1a_project_co... (Conv2D)	(None, 150, 150, 24)	960	block1a_se_excit...
block1a_project_bn (BatchNormalizatio...	(None, 150, 150, 24)	96	block1a_project_...
block1b_dwconv (DepthwiseConv2D)	(None, 150, 150, 24)	216	block1a_project_...
block1b_bn (BatchNormalizatio...	(None, 150, 150, 24)	96	block1b_dwconv[0...
block1b_activation (Activation)	(None, 150, 150, 24)	0	block1b_bn[0][0]

block1b_se_squeeze (GlobalAveragePool...)	(None, 24)	0	block1b_activati...
block1b_se_reshape (Reshape)	(None, 1, 1, 24)	0	block1b_se_squee...
block1b_se_reduce (Conv2D)	(None, 1, 1, 6)	150	block1b_se_resha...
block1b_se_expand (Conv2D)	(None, 1, 1, 24)	168	block1b_se_reduc...
block1b_se_excite (Multiply)	(None, 150, 150, 24)	0	block1b_activati... block1b_se_expan...
block1b_project_co... (Conv2D)	(None, 150, 150, 24)	576	block1b_se_excit...
block1b_project_bn (BatchNormalizatio...)	(None, 150, 150, 24)	96	block1b_project_...
block1b_drop (Dropout)	(None, 150, 150, 24)	0	block1b_project_...
block1b_add (Add)	(None, 150, 150, 24)	0	block1b_drop[0][... block1a_project_...
block2a_expand_conv (Conv2D)	(None, 150, 150, 144)	3,456	block1b_add[0][0]
block2a_expand_bn (BatchNormalizatio...)	(None, 150, 150, 144)	576	block2a_expand_c...
block2a_expand_act... (Activation)	(None, 150, 150, 144)	0	block2a_expand_b...
block2a_dwconv_pad (ZeroPadding2D)	(None, 151, 151, 144)	0	block2a_expand_a...
block2a_dwconv (DepthwiseConv2D)	(None, 75, 75, 144)	1,296	block2a_dwconv_p...
block2a_bn (BatchNormalizatio...)	(None, 75, 75, 144)	576	block2a_dwconv[0...
block2a_activation (Activation)	(None, 75, 75, 144)	0	block2a_bn[0][0]
block2a_se_squeeze (GlobalAveragePool...)	(None, 144)	0	block2a_activati...
block2a_se_reshape (Reshape)	(None, 1, 1, 144)	0	block2a_se_squee...
block2a_se_reduce (Conv2D)	(None, 1, 1, 6)	870	block2a_se_resha...
block2a_se_expand (Conv2D)	(None, 1, 1, 144)	1,008	block2a_se_reduc...
block2a_se_excite (Multiply)	(None, 75, 75, 144)	0	block2a_activati... block2a_se_expan...
block2a_project_co... (Conv2D)	(None, 75, 75, 32)	4,608	block2a_se_excit...
block2a_project_bn (BatchNormalizatio...)	(None, 75, 75, 32)	128	block2a_project_...

(BatchNormalization...)	32)		
block2b_expand_conv (Conv2D)	(None, 75, 75, 192)	6,144	block2a_project_...
block2b_expand_bn (BatchNormalization...)	(None, 75, 75, 192)	768	block2b_expand_c...
block2b_expand_act... (Activation)	(None, 75, 75, 192)	0	block2b_expand_b...
block2b_dwconv (DepthwiseConv2D)	(None, 75, 75, 192)	1,728	block2b_expand_a...
block2b_bn (BatchNormalization...)	(None, 75, 75, 192)	768	block2b_dwconv[0...
block2b_activation (Activation)	(None, 75, 75, 192)	0	block2b_bn[0][0]
block2b_se_squeeze (GlobalAveragePool...)	(None, 192)	0	block2b_activati...
block2b_se_reshape (Reshape)	(None, 1, 1, 192)	0	block2b_se_squee...
block2b_se_reduce (Conv2D)	(None, 1, 1, 8)	1,544	block2b_se_resha...
block2b_se_expand (Conv2D)	(None, 1, 1, 192)	1,728	block2b_se_reduc...
block2b_se_excite (Multiply)	(None, 75, 75, 192)	0	block2b_activati... block2b_se_expan...
block2b_project_co... (Conv2D)	(None, 75, 75, 32)	6,144	block2b_se_excit...
block2b_project_bn (BatchNormalization...)	(None, 75, 75, 32)	128	block2b_project_...
block2b_drop (Dropout)	(None, 75, 75, 32)	0	block2b_project_...
block2b_add (Add)	(None, 75, 75, 32)	0	block2b_drop[0][... block2a_project_...
block2c_expand_conv (Conv2D)	(None, 75, 75, 192)	6,144	block2b_add[0][0]
block2c_expand_bn (BatchNormalization...)	(None, 75, 75, 192)	768	block2c_expand_c...
block2c_expand_act... (Activation)	(None, 75, 75, 192)	0	block2c_expand_b...
block2c_dwconv (DepthwiseConv2D)	(None, 75, 75, 192)	1,728	block2c_expand_a...
block2c_bn (BatchNormalization...)	(None, 75, 75, 192)	768	block2c_dwconv[0...
block2c_activation (Activation)	(None, 75, 75, 192)	0	block2c_bn[0][0]
block2c_se_squeeze (GlobalAveragePool...)	(None, 192)	0	block2c_activati...

block2c_se_reshape (Reshape)	(None, 1, 1, 192)	0	block2c_se_squee...
block2c_se_reduce (Conv2D)	(None, 1, 1, 8)	1,544	block2c_se_resha...
block2c_se_expand (Conv2D)	(None, 1, 1, 192)	1,728	block2c_se_reduc...
block2c_se_excite (Multiply)	(None, 75, 75, 192)	0	block2c_activati... block2c_se_expan...
block2c_project_co... (Conv2D)	(None, 75, 75, 32)	6,144	block2c_se_excit...
block2c_project_bn (BatchNormalizatio...	(None, 75, 75, 32)	128	block2c_project_...
block2c_drop (Dropout)	(None, 75, 75, 32)	0	block2c_project_...
block2c_add (Add)	(None, 75, 75, 32)	0	block2c_drop[0][... block2b_add[0][0]
block3a_expand_conv (Conv2D)	(None, 75, 75, 192)	6,144	block2c_add[0][0]
block3a_expand_bn (BatchNormalizatio...	(None, 75, 75, 192)	768	block3a_expand_c...
block3a_expand_act... (Activation)	(None, 75, 75, 192)	0	block3a_expand_b...
block3a_dwconv_pad (ZeroPadding2D)	(None, 79, 79, 192)	0	block3a_expand_a...
block3a_dwconv (DepthwiseConv2D)	(None, 38, 38, 192)	4,800	block3a_dwconv_p...
block3a_bn (BatchNormalizatio...	(None, 38, 38, 192)	768	block3a_dwconv[0...
block3a_activation (Activation)	(None, 38, 38, 192)	0	block3a_bn[0][0]
block3a_se_squeeze (GlobalAveragePool...	(None, 192)	0	block3a_activati...
block3a_se_reshape (Reshape)	(None, 1, 1, 192)	0	block3a_se_squee...
block3a_se_reduce (Conv2D)	(None, 1, 1, 8)	1,544	block3a_se_resha...
block3a_se_expand (Conv2D)	(None, 1, 1, 192)	1,728	block3a_se_reduc...
block3a_se_excite (Multiply)	(None, 38, 38, 192)	0	block3a_activati... block3a_se_expan...
block3a_project_co... (Conv2D)	(None, 38, 38, 48)	9,216	block3a_se_excit...
block3a_project_bn (BatchNormalizatio...	(None, 38, 38, 48)	192	block3a_project_...
block3b_expand_conv	(None, 38, 38,	13,824	block3a_project_...

(Conv2D)	288)		
block3b_expand_bn (BatchNormalizatio...	(None, 38, 38, 288)	1,152	block3b_expand_c...
block3b_expand_act... (Activation)	(None, 38, 38, 288)	0	block3b_expand_b...
block3b_dwconv (DepthwiseConv2D)	(None, 38, 38, 288)	7,200	block3b_expand_a...
block3b_bn (BatchNormalizatio...	(None, 38, 38, 288)	1,152	block3b_dwconv[0...
block3b_activation (Activation)	(None, 38, 38, 288)	0	block3b_bn[0][0]
block3b_se_squeeze (GlobalAveragePool...	(None, 288)	0	block3b_activati...
block3b_se_reshape (Reshape)	(None, 1, 1, 288)	0	block3b_se_squee...
block3b_se_reduce (Conv2D)	(None, 1, 1, 12)	3,468	block3b_se_resha...
block3b_se_expand (Conv2D)	(None, 1, 1, 288)	3,744	block3b_se_reduc...
block3b_se_excite (Multiply)	(None, 38, 38, 288)	0	block3b_activati... block3b_se_expan...
block3b_project_co... (Conv2D)	(None, 38, 38, 48)	13,824	block3b_se_excit...
block3b_project_bn (BatchNormalizatio...	(None, 38, 38, 48)	192	block3b_project_...
block3b_drop (Dropout)	(None, 38, 38, 48)	0	block3b_project_...
block3b_add (Add)	(None, 38, 38, 48)	0	block3b_drop[0][... block3a_project_...
block3c_expand_conv (Conv2D)	(None, 38, 38, 288)	13,824	block3b_add[0][0]
block3c_expand_bn (BatchNormalizatio...	(None, 38, 38, 288)	1,152	block3c_expand_c...
block3c_expand_act... (Activation)	(None, 38, 38, 288)	0	block3c_expand_b...
block3c_dwconv (DepthwiseConv2D)	(None, 38, 38, 288)	7,200	block3c_expand_a...
block3c_bn (BatchNormalizatio...	(None, 38, 38, 288)	1,152	block3c_dwconv[0...
block3c_activation (Activation)	(None, 38, 38, 288)	0	block3c_bn[0][0]
block3c_se_squeeze (GlobalAveragePool...	(None, 288)	0	block3c_activati...
block3c_se_reshape (Reshape)	(None, 1, 1, 288)	0	block3c_se_squee...

block3c_se_reduce (Conv2D)	(None, 1, 1, 12)	3,468	block3c_se_resha...
block3c_se_expand (Conv2D)	(None, 1, 1, 288)	3,744	block3c_se_reduc...
block3c_se_excite (Multiply)	(None, 38, 38, 288)	0	block3c_activati... block3c_se_expan...
block3c_project_co... (Conv2D)	(None, 38, 38, 48)	13,824	block3c_se_excit...
block3c_project_bn (BatchNormalizatio...)	(None, 38, 38, 48)	192	block3c_project_...
block3c_drop (Dropout)	(None, 38, 38, 48)	0	block3c_project_...
block3c_add (Add)	(None, 38, 38, 48)	0	block3c_drop[0][...] block3b_add[0][0]
block4a_expand_conv (Conv2D)	(None, 38, 38, 288)	13,824	block3c_add[0][0]
block4a_expand_bn (BatchNormalizatio...)	(None, 38, 38, 288)	1,152	block4a_expand_c...
block4a_expand_act... (Activation)	(None, 38, 38, 288)	0	block4a_expand_b...
block4a_dwconv_pad (ZeroPadding2D)	(None, 39, 39, 288)	0	block4a_expand_a...
block4a_dwconv (DepthwiseConv2D)	(None, 19, 19, 288)	2,592	block4a_dwconv_p...
block4a_bn (BatchNormalizatio...)	(None, 19, 19, 288)	1,152	block4a_dwconv[0...
block4a_activation (Activation)	(None, 19, 19, 288)	0	block4a_bn[0][0]
block4a_se_squeeze (GlobalAveragePool...)	(None, 288)	0	block4a_activati...
block4a_se_reshape (Reshape)	(None, 1, 1, 288)	0	block4a_se_squee...
block4a_se_reduce (Conv2D)	(None, 1, 1, 12)	3,468	block4a_se_resha...
block4a_se_expand (Conv2D)	(None, 1, 1, 288)	3,744	block4a_se_reduc...
block4a_se_excite (Multiply)	(None, 19, 19, 288)	0	block4a_activati... block4a_se_expan...
block4a_project_co... (Conv2D)	(None, 19, 19, 96)	27,648	block4a_se_excit...
block4a_project_bn (BatchNormalizatio...)	(None, 19, 19, 96)	384	block4a_project_...
block4b_expand_conv (Conv2D)	(None, 19, 19, 576)	55,296	block4a_project_...
block4b_expand_bn	(None, 19, 19,	2,304	block4b_expand_c...

(BatchNormalizatio...	576)		
block4b_expand_act... (Activation)	(None, 19, 19, 576)	0	block4b_expand_b...
block4b_dwconv (DepthwiseConv2D)	(None, 19, 19, 576)	5,184	block4b_expand_a...
block4b_bn (BatchNormalizatio...	(None, 19, 19, 576)	2,304	block4b_dwconv[0...
block4b_activation (Activation)	(None, 19, 19, 576)	0	block4b_bn[0][0]
block4b_se_squeeze (GlobalAveragePool...	(None, 576)	0	block4b_activati...
block4b_se_reshape (Reshape)	(None, 1, 1, 576)	0	block4b_se_squee...
block4b_se_reduce (Conv2D)	(None, 1, 1, 24)	13,848	block4b_se_resha...
block4b_se_expand (Conv2D)	(None, 1, 1, 576)	14,400	block4b_se_reduc...
block4b_se_excite (Multiply)	(None, 19, 19, 576)	0	block4b_activati... block4b_se_expan...
block4b_project_co... (Conv2D)	(None, 19, 19, 96)	55,296	block4b_se_excit...
block4b_project_bn (BatchNormalizatio...	(None, 19, 19, 96)	384	block4b_project_...
block4b_drop (Dropout)	(None, 19, 19, 96)	0	block4b_project_...
block4b_add (Add)	(None, 19, 19, 96)	0	block4b_drop[0][... block4a_project_...
block4c_expand_conv (Conv2D)	(None, 19, 19, 576)	55,296	block4b_add[0][0]
block4c_expand_bn (BatchNormalizatio...	(None, 19, 19, 576)	2,304	block4c_expand_c...
block4c_expand_act... (Activation)	(None, 19, 19, 576)	0	block4c_expand_b...
block4c_dwconv (DepthwiseConv2D)	(None, 19, 19, 576)	5,184	block4c_expand_a...
block4c_bn (BatchNormalizatio...	(None, 19, 19, 576)	2,304	block4c_dwconv[0...
block4c_activation (Activation)	(None, 19, 19, 576)	0	block4c_bn[0][0]
block4c_se_squeeze (GlobalAveragePool...	(None, 576)	0	block4c_activati...
block4c_se_reshape (Reshape)	(None, 1, 1, 576)	0	block4c_se_squee...
block4c_se_reduce (Conv2D)	(None, 1, 1, 24)	13,848	block4c_se_resha...

block4c_se_expand (Conv2D)	(None, 1, 1, 576)	14,400	block4c_se_reduc...
block4c_se_excite (Multiply)	(None, 19, 19, 576)	0	block4c_activati... block4c_se_expan...
block4c_project_co... (Conv2D)	(None, 19, 19, 96)	55,296	block4c_se_excit...
block4c_project_bn (BatchNormalizatio...	(None, 19, 19, 96)	384	block4c_project_...
block4c_drop (Dropout)	(None, 19, 19, 96)	0	block4c_project_...
block4c_add (Add)	(None, 19, 19, 96)	0	block4c_drop[0][...] block4b_add[0][0]
block4d_expand_conv (Conv2D)	(None, 19, 19, 576)	55,296	block4c_add[0][0]
block4d_expand_bn (BatchNormalizatio...	(None, 19, 19, 576)	2,304	block4d_expand_c...
block4d_expand_act... (Activation)	(None, 19, 19, 576)	0	block4d_expand_b...
block4d_dwconv (DepthwiseConv2D)	(None, 19, 19, 576)	5,184	block4d_expand_a...
block4d_bn (BatchNormalizatio...	(None, 19, 19, 576)	2,304	block4d_dwconv[0...
block4d_activation (Activation)	(None, 19, 19, 576)	0	block4d_bn[0][0]
block4d_se_squeeze (GlobalAveragePool...	(None, 576)	0	block4d_activati...
block4d_se_reshape (Reshape)	(None, 1, 1, 576)	0	block4d_se_squee...
block4d_se_reduce (Conv2D)	(None, 1, 1, 24)	13,848	block4d_se_resha...
block4d_se_expand (Conv2D)	(None, 1, 1, 576)	14,400	block4d_se_reduc...
block4d_se_excite (Multiply)	(None, 19, 19, 576)	0	block4d_activati... block4d_se_expan...
block4d_project_co... (Conv2D)	(None, 19, 19, 96)	55,296	block4d_se_excit...
block4d_project_bn (BatchNormalizatio...	(None, 19, 19, 96)	384	block4d_project_...
block4d_drop (Dropout)	(None, 19, 19, 96)	0	block4d_project_...
block4d_add (Add)	(None, 19, 19, 96)	0	block4d_drop[0][...] block4c_add[0][0]
block4e_expand_conv (Conv2D)	(None, 19, 19, 576)	55,296	block4d_add[0][0]
block4e_expand_bn	(None, 19, 19,	2,304	block4e_expand_c...

(BatchNormalizatio...	576)		
block4e_expand_act... (Activation)	(None, 19, 19, 576)	0	block4e_expand_b...
block4e_dwconv (DepthwiseConv2D)	(None, 19, 19, 576)	5,184	block4e_expand_a...
block4e_bn (BatchNormalizatio...	(None, 19, 19, 576)	2,304	block4e_dwconv[0...
block4e_activation (Activation)	(None, 19, 19, 576)	0	block4e_bn[0][0]
block4e_se_squeeze (GlobalAveragePool...	(None, 576)	0	block4e_activati...
block4e_se_reshape (Reshape)	(None, 1, 1, 576)	0	block4e_se_squee...
block4e_se_reduce (Conv2D)	(None, 1, 1, 24)	13,848	block4e_se_resha...
block4e_se_expand (Conv2D)	(None, 1, 1, 576)	14,400	block4e_se_reduc...
block4e_se_excite (Multiply)	(None, 19, 19, 576)	0	block4e_activati... block4e_se_expan...
block4e_project_co... (Conv2D)	(None, 19, 19, 96)	55,296	block4e_se_excit...
block4e_project_bn (BatchNormalizatio...	(None, 19, 19, 96)	384	block4e_project_...
block4e_drop (Dropout)	(None, 19, 19, 96)	0	block4e_project_...
block4e_add (Add)	(None, 19, 19, 96)	0	block4e_drop[0][... block4d_add[0][0]
block5a_expand_conv (Conv2D)	(None, 19, 19, 576)	55,296	block4e_add[0][0]
block5a_expand_bn (BatchNormalizatio...	(None, 19, 19, 576)	2,304	block5a_expand_c...
block5a_expand_act... (Activation)	(None, 19, 19, 576)	0	block5a_expand_b...
block5a_dwconv (DepthwiseConv2D)	(None, 19, 19, 576)	14,400	block5a_expand_a...
block5a_bn (BatchNormalizatio...	(None, 19, 19, 576)	2,304	block5a_dwconv[0...
block5a_activation (Activation)	(None, 19, 19, 576)	0	block5a_bn[0][0]
block5a_se_squeeze (GlobalAveragePool...	(None, 576)	0	block5a_activati...
block5a_se_reshape (Reshape)	(None, 1, 1, 576)	0	block5a_se_squee...
block5a_se_reduce (Conv2D)	(None, 1, 1, 24)	13,848	block5a_se_resha...

block5a_se_expand (Conv2D)	(None, 1, 1, 576)	14,400	block5a_se_reduc...
block5a_se_excite (Multiply)	(None, 19, 19, 576)	0	block5a_activati... block5a_se_expan...
block5a_project_co... (Conv2D)	(None, 19, 19, 136)	78,336	block5a_se_excit...
block5a_project_bn (BatchNormalizatio...)	(None, 19, 19, 136)	544	block5a_project_...
block5b_expand_conv (Conv2D)	(None, 19, 19, 816)	110,976	block5a_project_...
block5b_expand_bn (BatchNormalizatio...)	(None, 19, 19, 816)	3,264	block5b_expand_c...
block5b_expand_act... (Activation)	(None, 19, 19, 816)	0	block5b_expand_b...
block5b_dwconv (DepthwiseConv2D)	(None, 19, 19, 816)	20,400	block5b_expand_a...
block5b_bn (BatchNormalizatio...)	(None, 19, 19, 816)	3,264	block5b_dwconv[0...
block5b_activation (Activation)	(None, 19, 19, 816)	0	block5b_bn[0][0]
block5b_se_squeeze (GlobalAveragePool...)	(None, 816)	0	block5b_activati...
block5b_se_reshape (Reshape)	(None, 1, 1, 816)	0	block5b_se_squee...
block5b_se_reduce (Conv2D)	(None, 1, 1, 34)	27,778	block5b_se_resha...
block5b_se_expand (Conv2D)	(None, 1, 1, 816)	28,560	block5b_se_reduc...
block5b_se_excite (Multiply)	(None, 19, 19, 816)	0	block5b_activati... block5b_se_expan...
block5b_project_co... (Conv2D)	(None, 19, 19, 136)	110,976	block5b_se_excit...
block5b_project_bn (BatchNormalizatio...)	(None, 19, 19, 136)	544	block5b_project_...
block5b_drop (Dropout)	(None, 19, 19, 136)	0	block5b_project_...
block5b_add (Add)	(None, 19, 19, 136)	0	block5b_drop[0][... block5a_project_...
block5c_expand_conv (Conv2D)	(None, 19, 19, 816)	110,976	block5b_add[0][0]
block5c_expand_bn (BatchNormalizatio...)	(None, 19, 19, 816)	3,264	block5c_expand_c...
block5c_expand_act... (Activation)	(None, 19, 19, 816)	0	block5c_expand_b...
block5c_dwconv	(None, 19, 19, 816)	20,400	block5c_expand a...

block5c_dwconv (DepthwiseConv2D)	(None, 19, 19, 816)	3,264	block5c_dwconv[0][0]
block5c_bn (BatchNormalization)	(None, 19, 19, 816)	0	block5c_bn[0][0]
block5c_activation (Activation)	(None, 19, 19, 816)	0	block5c_activation[0][0]
block5c_se_squeeze (GlobalAveragePooling2D)	(None, 816)	0	block5c_se_squeeze[0][0]
block5c_se_reshape (Reshape)	(None, 1, 1, 816)	0	block5c_se_reshape[0][0]
block5c_se_reduce (Conv2D)	(None, 1, 1, 34)	27,778	block5c_se_reduce[0][0]
block5c_se_expand (Conv2D)	(None, 1, 1, 816)	28,560	block5c_se_expand[0][0]
block5c_se_excite (Multiply)	(None, 19, 19, 816)	0	block5c_se_excite[0][0]
block5c_project_conv (Conv2D)	(None, 19, 19, 136)	110,976	block5c_project_conv[0][0]
block5c_project_bn (BatchNormalization)	(None, 19, 19, 136)	544	block5c_project_bn[0][0]
block5c_drop (Dropout)	(None, 19, 19, 136)	0	block5c_drop[0][0]
block5c_add (Add)	(None, 19, 19, 136)	0	block5c_add[0][0]
block5d_expand_conv (Conv2D)	(None, 19, 19, 816)	110,976	block5d_expand_conv[0][0]
block5d_expand_bn (BatchNormalization)	(None, 19, 19, 816)	3,264	block5d_expand_bn[0][0]
block5d_expand_activation (Activation)	(None, 19, 19, 816)	0	block5d_expand_activation[0][0]
block5d_dwconv (DepthwiseConv2D)	(None, 19, 19, 816)	20,400	block5d_dwconv[0][0]
block5d_bn (BatchNormalization)	(None, 19, 19, 816)	3,264	block5d_bn[0][0]
block5d_activation (Activation)	(None, 19, 19, 816)	0	block5d_activation[0][0]
block5d_se_squeeze (GlobalAveragePooling2D)	(None, 816)	0	block5d_se_squeeze[0][0]
block5d_se_reshape (Reshape)	(None, 1, 1, 816)	0	block5d_se_reshape[0][0]
block5d_se_reduce (Conv2D)	(None, 1, 1, 34)	27,778	block5d_se_reduce[0][0]
block5d_se_expand (Conv2D)	(None, 1, 1, 816)	28,560	block5d_se_expand[0][0]
block5d_se_excite (Multiply)	(None, 19, 19, 816)	0	block5d_se_excite[0][0]

block5d_project_co... (Conv2D)	(None, 19, 19, 136)	110,976	block5d_se_excit...
block5d_project_bn (BatchNormalizatio...	(None, 19, 19, 136)	544	block5d_project_...
block5d_drop (Dropout)	(None, 19, 19, 136)	0	block5d_project_...
block5d_add (Add)	(None, 19, 19, 136)	0	block5d_drop[0][... block5c_add[0][0]
block5e_expand_conv (Conv2D)	(None, 19, 19, 816)	110,976	block5d_add[0][0]
block5e_expand_bn (BatchNormalizatio...	(None, 19, 19, 816)	3,264	block5e_expand_c...
block5e_expand_act... (Activation)	(None, 19, 19, 816)	0	block5e_expand_b...
block5e_dwconv (DepthwiseConv2D)	(None, 19, 19, 816)	20,400	block5e_expand_a...
block5e_bn (BatchNormalizatio...	(None, 19, 19, 816)	3,264	block5e_dwconv[0...
block5e_activation (Activation)	(None, 19, 19, 816)	0	block5e_bn[0][0]
block5e_se_squeeze (GlobalAveragePool...	(None, 816)	0	block5e_activati...
block5e_se_reshape (Reshape)	(None, 1, 1, 816)	0	block5e_se_squee...
block5e_se_reduce (Conv2D)	(None, 1, 1, 34)	27,778	block5e_se_resha...
block5e_se_expand (Conv2D)	(None, 1, 1, 816)	28,560	block5e_se_reduc...
block5e_se_excite (Multiply)	(None, 19, 19, 816)	0	block5e_activati... block5e_se_expan...
block5e_project_co... (Conv2D)	(None, 19, 19, 136)	110,976	block5e_se_excit...
block5e_project_bn (BatchNormalizatio...	(None, 19, 19, 136)	544	block5e_project_...
block5e_drop (Dropout)	(None, 19, 19, 136)	0	block5e_project_...
block5e_add (Add)	(None, 19, 19, 136)	0	block5e_drop[0][... block5d_add[0][0]
block6a_expand_conv (Conv2D)	(None, 19, 19, 816)	110,976	block5e_add[0][0]
block6a_expand_bn (BatchNormalizatio...	(None, 19, 19, 816)	3,264	block6a_expand_c...
block6a_expand_act... (Activation)	(None, 19, 19, 816)	0	block6a_expand_b...
block6a_dwconv_pad	(None, 22, 22, 816)	0	block6a_expand_a...

block6a_dwconv_pau (ZeroPadding2D)	(None, 23, 23, 816)	0	block6a_expand_a...
block6a_dwconv (DepthwiseConv2D)	(None, 10, 10, 816)	20,400	block6a_dwconv_p...
block6a_bn (BatchNormalizatio...	(None, 10, 10, 816)	3,264	block6a_dwconv[0...
block6a_activation (Activation)	(None, 10, 10, 816)	0	block6a_bn[0][0]
block6a_se_squeeze (GlobalAveragePool...	(None, 816)	0	block6a_activati...
block6a_se_reshape (Reshape)	(None, 1, 1, 816)	0	block6a_se_squee...
block6a_se_reduce (Conv2D)	(None, 1, 1, 34)	27,778	block6a_se_resha...
block6a_se_expand (Conv2D)	(None, 1, 1, 816)	28,560	block6a_se_reduc...
block6a_se_excite (Multiply)	(None, 10, 10, 816)	0	block6a_activati... block6a_se_expan...
block6a_project_co... (Conv2D)	(None, 10, 10, 232)	189,312	block6a_se_excit...
block6a_project_bn (BatchNormalizatio...	(None, 10, 10, 232)	928	block6a_project_...
block6b_expand_conv (Conv2D)	(None, 10, 10, 1392)	322,944	block6a_project_...
block6b_expand_bn (BatchNormalizatio...	(None, 10, 10, 1392)	5,568	block6b_expand_c...
block6b_expand_act... (Activation)	(None, 10, 10, 1392)	0	block6b_expand_b...
block6b_dwconv (DepthwiseConv2D)	(None, 10, 10, 1392)	34,800	block6b_expand_a...
block6b_bn (BatchNormalizatio...	(None, 10, 10, 1392)	5,568	block6b_dwconv[0...
block6b_activation (Activation)	(None, 10, 10, 1392)	0	block6b_bn[0][0]
block6b_se_squeeze (GlobalAveragePool...	(None, 1392)	0	block6b_activati...
block6b_se_reshape (Reshape)	(None, 1, 1, 1392)	0	block6b_se_squee...
block6b_se_reduce (Conv2D)	(None, 1, 1, 58)	80,794	block6b_se_resha...
block6b_se_expand (Conv2D)	(None, 1, 1, 1392)	82,128	block6b_se_reduc...
block6b_se_excite (Multiply)	(None, 10, 10, 1392)	0	block6b_activati... block6b_se_expan...
block6b_project_co... (Conv2D)	(None, 10, 10, 232)	322,944	block6b_se_excit...

block6b_project_bn (BatchNormalizatio...	(None, 10, 10, 232)	928	block6b_project_...
block6b_drop (Dropout)	(None, 10, 10, 232)	0	block6b_project_...
block6b_add (Add)	(None, 10, 10, 232)	0	block6b_drop[0][... block6a_project_...
block6c_expand_conv (Conv2D)	(None, 10, 10, 1392)	322,944	block6b_add[0][0]
block6c_expand_bn (BatchNormalizatio...	(None, 10, 10, 1392)	5,568	block6c_expand_c...
block6c_expand_act... (Activation)	(None, 10, 10, 1392)	0	block6c_expand_b...
block6c_dwconv (DepthwiseConv2D)	(None, 10, 10, 1392)	34,800	block6c_expand_a...
block6c_bn (BatchNormalizatio...	(None, 10, 10, 1392)	5,568	block6c_dwconv[0...
block6c_activation (Activation)	(None, 10, 10, 1392)	0	block6c_bn[0][0]
block6c_se_squeeze (GlobalAveragePool...	(None, 1392)	0	block6c_activati...
block6c_se_reshape (Reshape)	(None, 1, 1, 1392)	0	block6c_se_squee...
block6c_se_reduce (Conv2D)	(None, 1, 1, 58)	80,794	block6c_se_resha...
block6c_se_expand (Conv2D)	(None, 1, 1, 1392)	82,128	block6c_se_reduc...
block6c_se_excite (Multiply)	(None, 10, 10, 1392)	0	block6c_activati... block6c_se_expan...
block6c_project_co... (Conv2D)	(None, 10, 10, 232)	322,944	block6c_se_excit...
block6c_project_bn (BatchNormalizatio...	(None, 10, 10, 232)	928	block6c_project_...
block6c_drop (Dropout)	(None, 10, 10, 232)	0	block6c_project_...
block6c_add (Add)	(None, 10, 10, 232)	0	block6c_drop[0][... block6b_add[0][0]
block6d_expand_conv (Conv2D)	(None, 10, 10, 1392)	322,944	block6c_add[0][0]
block6d_expand_bn (BatchNormalizatio...	(None, 10, 10, 1392)	5,568	block6d_expand_c...
block6d_expand_act... (Activation)	(None, 10, 10, 1392)	0	block6d_expand_b...
block6d_dwconv (DepthwiseConv2D)	(None, 10, 10, 1392)	34,800	block6d_expand_a...

block6d_bn (BatchNormalizatio...	(None, 10, 10, 1392)	5,568	block6d_dwconv[0...
block6d_activation (Activation)	(None, 10, 10, 1392)	0	block6d_bn[0][0]
block6d_se_squeeze (GlobalAveragePool...	(None, 1392)	0	block6d_activati...
block6d_se_reshape (Reshape)	(None, 1, 1, 1392)	0	block6d_se_squee...
block6d_se_reduce (Conv2D)	(None, 1, 1, 58)	80,794	block6d_se_resha...
block6d_se_expand (Conv2D)	(None, 1, 1, 1392)	82,128	block6d_se_reduc...
block6d_se_excite (Multiply)	(None, 10, 10, 1392)	0	block6d_activati... block6d_se_expan...
block6d_project_co... (Conv2D)	(None, 10, 10, 232)	322,944	block6d_se_excit...
block6d_project_bn (BatchNormalizatio...	(None, 10, 10, 232)	928	block6d_project_...
block6d_drop (Dropout)	(None, 10, 10, 232)	0	block6d_project_...
block6d_add (Add)	(None, 10, 10, 232)	0	block6d_drop[0][... block6c_add[0][0]
block6e_expand_conv (Conv2D)	(None, 10, 10, 1392)	322,944	block6d_add[0][0]
block6e_expand_bn (BatchNormalizatio...	(None, 10, 10, 1392)	5,568	block6e_expand_c...
block6e_expand_act... (Activation)	(None, 10, 10, 1392)	0	block6e_expand_b...
block6e_dwconv (DepthwiseConv2D)	(None, 10, 10, 1392)	34,800	block6e_expand_a...
block6e_bn (BatchNormalizatio...	(None, 10, 10, 1392)	5,568	block6e_dwconv[0...
block6e_activation (Activation)	(None, 10, 10, 1392)	0	block6e_bn[0][0]
block6e_se_squeeze (GlobalAveragePool...	(None, 1392)	0	block6e_activati...
block6e_se_reshape (Reshape)	(None, 1, 1, 1392)	0	block6e_se_squee...
block6e_se_reduce (Conv2D)	(None, 1, 1, 58)	80,794	block6e_se_resha...
block6e_se_expand (Conv2D)	(None, 1, 1, 1392)	82,128	block6e_se_reduc...
block6e_se_excite (Multiply)	(None, 10, 10, 1392)	0	block6e_activati... block6e_se_expan...
block6e_project_co... (Conv2D)	(None, 10, 10, 232)	322,944	block6e_se_excit...

block6e_project_bn (BatchNormalizatio...	(None, 10, 10, 232)	928	block6e_project_...
block6e_drop (Dropout)	(None, 10, 10, 232)	0	block6e_project_...
block6e_add (Add)	(None, 10, 10, 232)	0	block6e_drop[0][...] block6d_add[0][0]
block6f_expand_conv (Conv2D)	(None, 10, 10, 1392)	322,944	block6e_add[0][0]
block6f_expand_bn (BatchNormalizatio...	(None, 10, 10, 1392)	5,568	block6f_expand_c...
block6f_expand_act... (Activation)	(None, 10, 10, 1392)	0	block6f_expand_b...
block6f_dwconv (DepthwiseConv2D)	(None, 10, 10, 1392)	34,800	block6f_expand_a...
block6f_bn (BatchNormalizatio...	(None, 10, 10, 1392)	5,568	block6f_dwconv[0...
block6f_activation (Activation)	(None, 10, 10, 1392)	0	block6f_bn[0][0]
block6f_se_squeeze (GlobalAveragePool...	(None, 1392)	0	block6f_activati...
block6f_se_reshape (Reshape)	(None, 1, 1, 1392)	0	block6f_se_squee...
block6f_se_reduce (Conv2D)	(None, 1, 1, 58)	80,794	block6f_se_resha...
block6f_se_expand (Conv2D)	(None, 1, 1, 1392)	82,128	block6f_se_reduc...
block6f_se_excite (Multiply)	(None, 10, 10, 1392)	0	block6f_activati... block6f_se_expan...
block6f_project_co... (Conv2D)	(None, 10, 10, 232)	322,944	block6f_se_excit...
block6f_project_bn (BatchNormalizatio...	(None, 10, 10, 232)	928	block6f_project_...
block6f_drop (Dropout)	(None, 10, 10, 232)	0	block6f_project_...
block6f_add (Add)	(None, 10, 10, 232)	0	block6f_drop[0][...] block6e_add[0][0]
block7a_expand_conv (Conv2D)	(None, 10, 10, 1392)	322,944	block6f_add[0][0]
block7a_expand_bn (BatchNormalizatio...	(None, 10, 10, 1392)	5,568	block7a_expand_c...
block7a_expand_act... (Activation)	(None, 10, 10, 1392)	0	block7a_expand_b...
block7a_dwconv (DepthwiseConv2D)	(None, 10, 10, 1392)	12,528	block7a_expand_a...

block7a_bn	(None, 10, 10,	5,568	block7a_dwconv 0...
------------	----------------	-------	---------------------

```
model = load_model("efficientnet_korean_food.keras")
```

(Activation)	1392		
--------------	------	--	--

```
from tensorflow.keras.applications.efficientnet import preprocess_input
```

```
def preprocess(image, label):
    image = preprocess_input(image)
    return image, label
```

(Conv2D)			
----------	--	--	--

```
train_dataset = train_dataset.map(preprocess)
val_dataset = val_dataset.map(preprocess)
```

block7a_se_excite	(None, 10, 10,	0	block7a_activati...
-------------------	----------------	---	---------------------

```
from tensorflow.keras.optimizers import Adam
```

```
model.compile(
    optimizer=Adam(1e-5),
    loss="categorical_crossentropy",
    metrics=["accuracy"]
)
```

```
history = model.fit(
    train_dataset,
    validation_data=val_dataset,
    epochs=10
)
```

Epoch 1/10	block7b_dwconv	(None, 10, 10,	20,736	block7b_expand_a...	
125/125	batch normalization	(None, 10, 10,	2304		
Epoch 2/10	block7b_bn	(None, 10, 10,	584	block7b_dwconv	0.4596 - val_accuracy: 0.
125/125	batch normalization	(None, 10, 10,	2304		
Epoch 3/10	block7b_activation	(None, 10, 10,	0	block7b_bn[0][0]	0.9023 - loss: 0.4159 - val_accuracy: 0.
125/125	batch normalization	(None, 10, 10,	2304		
Epoch 4/10	block7b_se.squeeze	(None, 10, 10,	615	block7b_activation	0.9202 - loss: 0.3601 - val_accuracy: 0.
125/125	GlobalAveragePool...	(None, 10, 10,	2304		
Epoch 5/10	block7b_se.reshape	(None, 1, 1,	0	block7b_se.squeeze	0.9210 - val_accuracy: 0.
125/125	Reshape	(None, 1, 1,	2304		
Epoch 6/10	block7b_se.reduce	(None, 1, 1, 96)	221,280	block7b_se.reshape	0.9315 - loss: 0.2895 - val_accuracy: 0.
125/125	Conv2D	(None, 1, 1,	223,488		
Epoch 7/10	block7b_se.expand	(None, 1, 1,	223,488	block7b_se.reduce	0.9379 - loss: 0.2615 - val_accuracy: 0.
125/125	Conv2D	(None, 1, 1,	223,488		
Epoch 8/10	block7b_se_excite	(None, 10, 10,	0	block7b_activati...	0.9514 - loss: 0.2265 - val_accuracy: 0.
125/125	batch normalization	(None, 10, 10,	2304		
Epoch 9/10	block7b_se.expand	(None, 1, 1,	223,488	block7b_se_reduce	0.9541 - loss: 0.2039 - val_accuracy: 0.
125/125	Conv2D	(None, 1, 1,	223,488		
Epoch 10/10	block7b_se_excite	(None, 10, 10,	0	block7b_activati...	0.9610 - loss: 0.1797 - val_accuracy: 0.
125/125	batch normalization	(None, 10, 10,	2304		

```
history.history["accuracy"][-1], history.history["val_accuracy"][-1]
```

0.9624999761581421, 0.9900000095367432)			
block7b_project_bn	(None, 10, 10,	1,536	block7b_project_...
(Conv2D)			

```
train_dataset = tf.keras.utils.image_dataset_from_directory(
    dataset_path,
```

```

        image_size=(224, 224),
        batch_size=32,
        validation_split=0.2,
        subset="training",
        seed=123
    )

```

```

    (BatchNormalization, 1536)
Found 5000 files belonging to 25 classes.
Using 4000 files for training.
top_activation (None, 10, 10, 0 top_bn[0][0]
(Activation) 1536)

```

```

test_dataset = tf.keras.utils.image_dataset_from_directory(
    dataset_path,
    image_size=(300, 300),
    batch_size=32,
    shuffle=False
)

```

```

Total params: 25,744,176 (257.40 MB)
Trainable params: 10,714,656 (106.25 MB)
Non-trainable params: 87,303 (341.03 KB)

```

```

test_dataset = tf.keras.utils.image_dataset_from_directory(
    dataset_path,
    image_size=(300, 300),
    batch_size=32,
    shuffle=False,
    label_mode="categorical" # ← WAJIB untuk model categorical_crossentropy
)

```

```

loss, acc = model.evaluate(test_dataset)
print("Akurasi keseluruhan:", acc)

```

```

Found 5000 files belonging to 25 classes.
157/157 ————— 19s 122ms/step - accuracy: 0.9852 - loss: 0.0864
Akurasi keseluruhan: 0.9869999885559082

```

```

model.save("EfficientNetB3_korean_food_final.keras")

```

```

from google.colab import files
files.download("EfficientNetB3_korean_food_final.keras")

```

```

import matplotlib.pyplot as plt

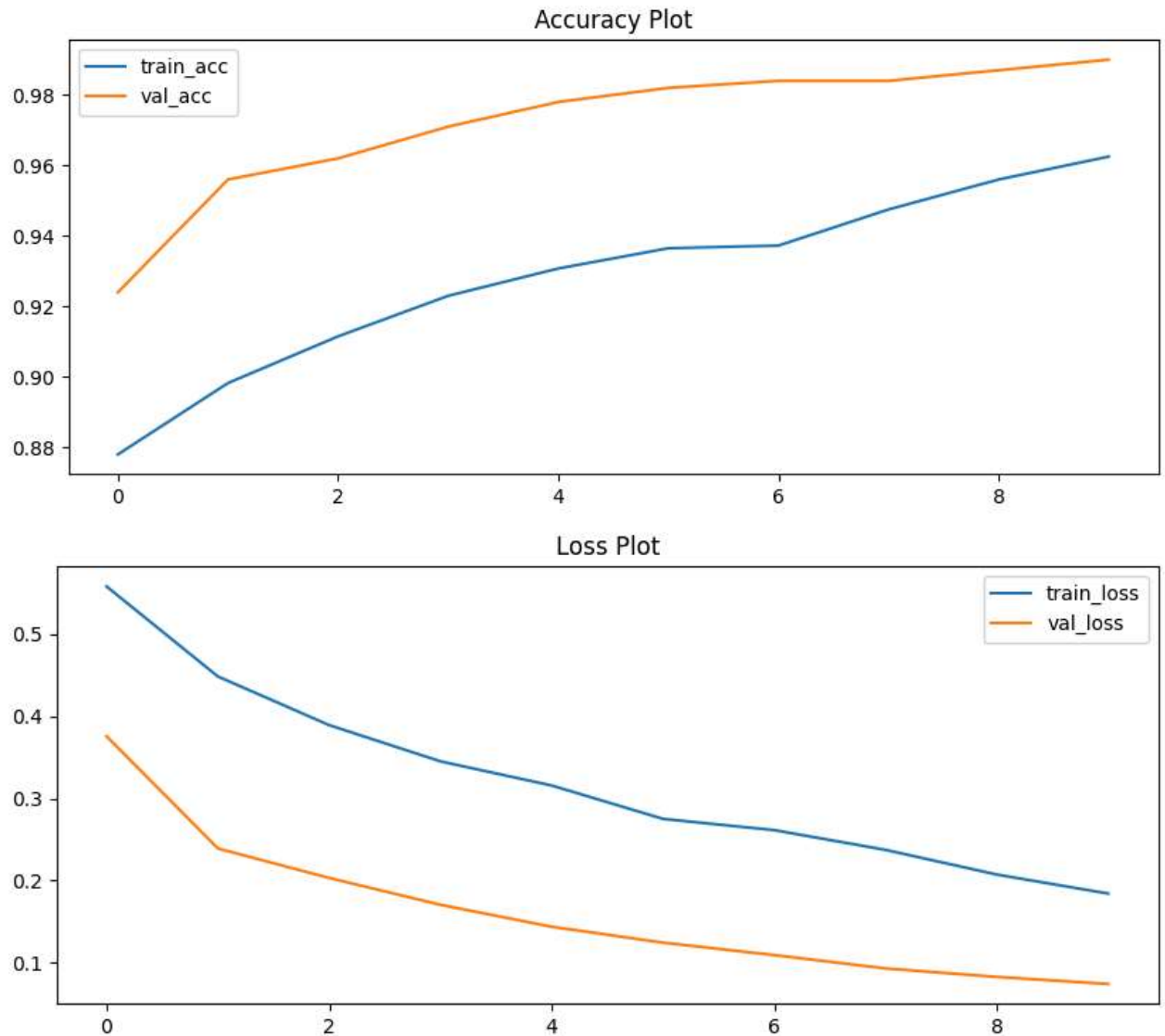
acc = history.history["accuracy"]
val_acc = history.history["val_accuracy"]
loss = history.history["loss"]
val_loss = history.history["val_loss"]

plt.figure(figsize=(10,4))
plt.plot(acc, label="train_acc")
plt.plot(val_acc, label="val_acc")
plt.legend()
plt.title("Accuracy Plot")

```

```
plt.show()

plt.figure(figsize=(10,4))
plt.plot(loss, label="train_loss")
plt.plot(val_loss, label="val_loss")
plt.legend()
plt.title("Loss Plot")
plt.show()
```



```
import numpy as np
from tensorflow.keras.preprocessing import image

img = image.load_img("/content/korean_food_dataset/content/korean_food_dataset/sushi/sushi_0.jpg",
                     target_size=(300,300))
img_array = image.img_to_array(img)
img_array = np.expand_dims(img_array, axis=0)

pred = model.predict(img_array)
class_index = np.argmax(pred)
```

```
print("Prediksi kelas:", class_index)
```

1/1 ————— 0s 44ms/step

Prediksi kelas: 23

```
import tensorflow as tf
import numpy as np
import matplotlib.pyplot as plt

# --- 1. Load class names ---
class_names = train_dataset.class_names # pastikan ini sesuai dataset kamu

# --- 2. Fungsi load & preprocess gambar ---
def load_and_preprocess(img_path):
    ... img = tf.keras.utils.load_img(img_path, target_size=(224, 224))
    ... img_array = tf.keras.utils.img_to_array(img)
    ... img_array = np.expand_dims(img_array, axis=0) # tambahkan batch dimension
    ... img_array = img_array / 255.0 # normalisasi
    ... return img, img_array

# --- 3. Prediksi ---
img_path = "/content/korean_food_dataset/content/korean_food_dataset/sushi/sushi_0.jpg" # ganti
img, img_array = load_and_preprocess(img_path)

pred = model.predict(img_array)
pred_class = class_names[np.argmax(pred)]
pred_prob = np.max(pred) * 100

# --- 4. Tampilkan gambar + hasil prediksi ---
plt.imshow(img)
plt.axis("off")
plt.title(f"Prediksi: {pred_class} ({pred_prob:.2f}%)")
plt.show()
```

1/1 ————— 17s 17s/step

Prediksi: lobster_bisque (24.55%)



```
import numpy as np

y_true = []
y_pred = []

for images, labels in test_dataset:
    preds = model.predict(images)
    y_true.extend(np.argmax(labels.numpy(), axis=1))
    y_pred.extend(np.argmax(preds, axis=1))

y_true = np.array(y_true)
y_pred = np.array(y_pred)
```

```
1/1 ██████████ 0s 336ms/step
1/1 ██████████ 0s 234ms/step
1/1 ██████████ 0s 251ms/step
1/1 ██████████ 0s 241ms/step
1/1 ██████████ 0s 232ms/step
1/1 ██████████ 0s 230ms/step
1/1 ██████████ 0s 204ms/step
1/1 ██████████ 0s 162ms/step
1/1 ██████████ 0s 187ms/step
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1/1 ██████████ 0s 162ms/step
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1/1 ██████████ 0s 182ms/step
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1/1 ██████████ 0s 185ms/step
1/1 ██████████ 0s 212ms/step
1/1 ██████████ 0s 197ms/step
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1/1 ██████████ 0s 163ms/step
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1/1 ██████████ 0s 176ms/step
1/1 ██████████ 0s 167ms/step
1/1 ██████████ 0s 185ms/step
```

```

1/1 _____ 0s 163ms/step
1/1 _____ 0s 187ms/step
1/1 _____ 0s 182ms/step
1/1 _____ 0s 180ms/step
1/1 _____ 0s 183ms/step
1/1 _____ 0s 190ms/step
1/1 _____ 0s 185ms/step
1/1 _____ 0s 166ms/step

```

```

from sklearn.metrics import confusion_matrix
cm = confusion_matrix(y_true, y_pred)

```

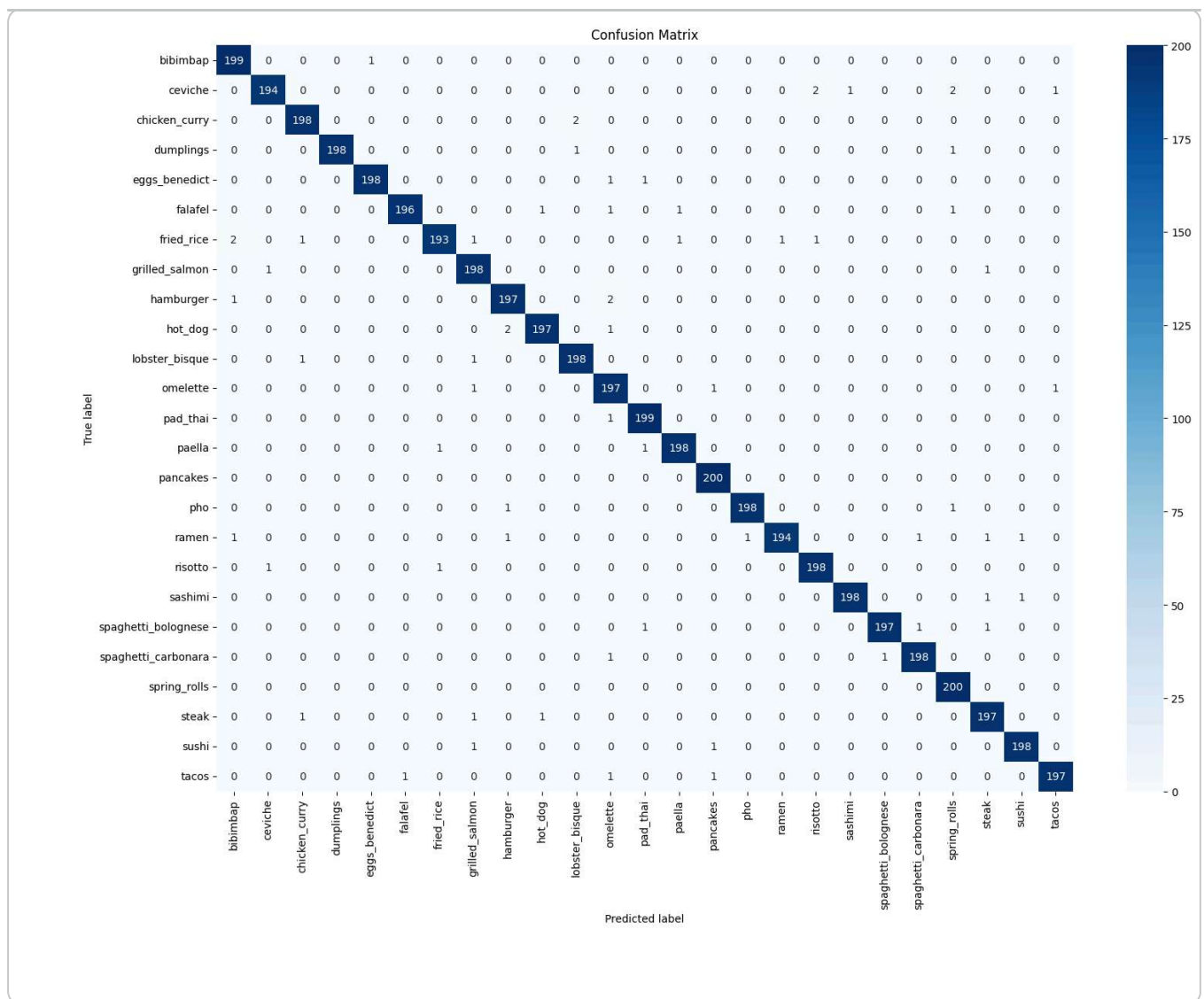
```

import matplotlib.pyplot as plt
import seaborn as sns

plt.figure(figsize=(16, 12))
sns.heatmap(
    cm,
    annot=True,          # tampilkan angka
    fmt="d",             # format angka integer
    cmap="Blues",        # warna biru
    xticklabels=class_names,
    yticklabels=class_names
)

plt.xlabel("Predicted label")
plt.ylabel("True label")
plt.title("Confusion Matrix")
plt.xticks(rotation=90)
plt.yticks(rotation=0)
plt.tight_layout()
plt.show()

```



```

food_labels = [
    "bibimbap",
    "ceviche",
    "chicken_curry",
    "dumplings",
    "eggs_benedict",
    "falafel",
    "fried_rice",
    "grilled_salmon",
    "hamburger",
    "hot_dog",
    "lobster_bisque",
    "omelette",
    "pad_thai",
    "paella",
    "pancakes",
    "pho",
    "ramen",
    "risotto",
    "sashimi",
    "spaghetti_bolognese",
    "spaghetti_carbonara",
    "spring_rolls",
    "steak",

```

```
"sushi",
"tacos"
]
```

```
from sklearn.metrics import classification_report

print(classification_report(y_true, y_pred, target_names=food_labels))
```

	precision	recall	f1-score	support
bibimbap	0.98	0.99	0.99	200
ceviche	0.99	0.97	0.98	200
chicken_curry	0.99	0.99	0.99	200
dumplings	1.00	0.99	0.99	200
eggs_benedict	0.99	0.99	0.99	200
falafel	0.99	0.98	0.99	200
fried_rice	0.99	0.96	0.98	200
grilled_salmon	0.98	0.99	0.98	200
hamburger	0.98	0.98	0.98	200
hot_dog	0.99	0.98	0.99	200
lobster_bisque	0.99	0.99	0.99	200
omelette	0.96	0.98	0.97	200
pad_thai	0.99	0.99	0.99	200
paella	0.99	0.99	0.99	200
pancakes	0.99	1.00	0.99	200
pho	0.99	0.99	0.99	200
ramen	0.99	0.97	0.98	200
risotto	0.99	0.99	0.99	200
sashimi	0.99	0.99	0.99	200
spaghetti_bolognese	0.99	0.98	0.99	200
spaghetti_carbonara	0.99	0.99	0.99	200
spring_rolls	0.98	1.00	0.99	200
steak	0.98	0.98	0.98	200
sushi	0.99	0.99	0.99	200
tacos	0.99	0.98	0.99	200
accuracy			0.99	5000
macro avg	0.99	0.99	0.99	5000
weighted avg	0.99	0.99	0.99	5000

```
import matplotlib.pyplot as plt
import numpy as np

# ----- Nama makanan sesuai urutan kelas -----
food_labels = [
    "bibimbap",
    "ceviche",
    "chicken_curry",
    "dumplings",
    "eggs_benedict",
    "falafel",
    "fried_rice",
    "grilled_salmon",
    "hamburger",
    "hot_dog",
    "lobster_bisque",
    "omelette",
    "pad_thai",
```

```

    "paella",
    "pancakes",
    "pho",
    "ramen",
    "risotto",
    "sashimi",
    "spaghetti_bolognese",
    "spaghetti_carbonara",
    "spring_rolls",
    "steak",
    "sushi",
    "tacos"
]

# ----- Precision, Recall, F1 -----
precision = [
    0.98,0.99,0.99,1.00,0.99,0.99,0.99,0.98,0.98,0.99,
    0.99,0.96,0.99,0.99,0.99,0.99,0.99,0.99,0.99,0.99,
    0.99,0.98,0.98,0.99,0.99
]

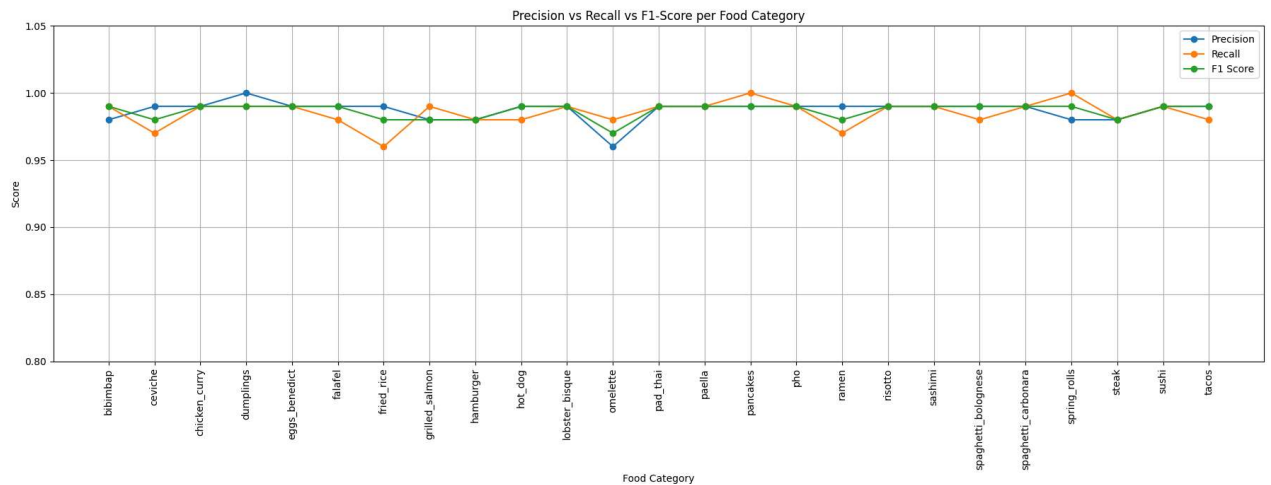
recall = [
    0.99,0.97,0.99,0.99,0.99,0.98,0.96,0.99,0.98,0.98,
    0.99,0.98,0.99,0.99,1.00,0.99,0.97,0.99,0.99,0.98,
    0.99,1.00,0.98,0.99,0.98
]

f1_score = [
    0.99,0.98,0.99,0.99,0.99,0.99,0.98,0.98,0.98,0.99,
    0.99,0.97,0.99,0.99,0.99,0.99,0.98,0.99,0.99,0.99,
    0.99,0.99,0.98,0.99,0.99
]

# ----- Plot -----
plt.figure(figsize=(18,7))
plt.plot(food_labels, precision, marker='o', label='Precision')
plt.plot(food_labels, recall, marker='o', label='Recall')
plt.plot(food_labels, f1_score, marker='o', label='F1 Score')

plt.title("Precision vs Recall vs F1-Score per Food Category")
plt.xlabel("Food Category")
plt.ylabel("Score")
plt.xticks(rotation=90)
plt.ylim(0.8, 1.05)
plt.grid(True)
plt.legend()
plt.tight_layout()
plt.show()

```



```
import tensorflow as tf
import numpy as np
import matplotlib.pyplot as plt
from tensorflow.keras.applications.efficientnet import preprocess_input

# --- Pakai label fix berdasarkan dataset ---
class_names = [
    "bibimbap", "ceviche", "chicken_curry", "dumplings", "eggs_benedict",
    "falafel", "fried_rice", "grilled_salmon", "hamburger", "hot_dog",
    "lobster_bisque", "omelette", "pad_thai", "paella", "pancakes",
    "pho", "ramen", "risotto", "sashimi", "spaghetti_bolognese",
    "spaghetti_carbonara", "spring_rolls", "steak", "sushi", "tacos"
]

def load_and_preprocess(img_path):
    img = tf.keras.utils.load_img(img_path, target_size=(224, 224))
    img_array = tf.keras.utils.img_to_array(img)
    img_array = np.expand_dims(img_array, axis=0)
    img_array = preprocess_input(img_array)  # WAJIB UNTUK EFFICIENTNET
    return img, img_array

img_path = "/content/korean_food_dataset/content/korean_food_dataset/dumplings/dumplings_109.jp
img, img_array = load_and_preprocess(img_path)

pred = model.predict(img_array)
pred_class = class_names[np.argmax(pred)]
pred_prob = np.max(pred) * 100

plt.imshow(img)
plt.axis("off")
plt.title(f"Prediksi: {pred_class} ({pred_prob:.2f}%)"
plt.show()
```

1/1 ————— 0s 69ms/step

Prediksi: dumplings (95.80%)



```
import tensorflow as tf
import numpy as np
import matplotlib.pyplot as plt
from tensorflow.keras.applications.efficientnet import preprocess_input

# --- Pakai label fix berdasarkan dataset ---
class_names = [
    "bibimbap", "ceviche", "chicken_curry", "dumplings", "eggs_benedict",
    "falafel", "fried_rice", "grilled_salmon", "hamburger", "hot_dog",
    "lobster_bisque", "omelette", "pad_thai", "paella", "pancakes",
    "pho", "ramen", "risotto", "sashimi", "spaghetti_bolognese",
    "spaghetti_carbonara", "spring_rolls", "steak", "sushi", "tacos"
]

def load_and_preprocess(img_path):
    img = tf.keras.utils.load_img(img_path, target_size=(224, 224))
    img_array = tf.keras.utils.img_to_array(img)
    img_array = np.expand_dims(img_array, axis=0)
    img_array = preprocess_input(img_array) # WAJIB UNTUK EFFICIENTNET
    return img, img_array

img_path = "/content/korean_food_dataset/content/korean_food_dataset/pho/pho_137.jpg"
img, img_array = load_and_preprocess(img_path)

pred = model.predict(img_array)
pred_class = class_names[np.argmax(pred)]
pred_prob = np.max(pred) * 100

plt.imshow(img)
plt.axis("off")
plt.title(f"Prediksi: {pred_class} ({pred_prob:.2f}%)")
plt.show()
```